

# Abdullah Hashmat

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## EDUCATION

**Lahore University of Management Science - BS Computer Science**

Aug 2021-May 2025

Relevant Courses: CS6304 Advanced Machine Learning, CS5302 Speech Processing with Generative AI, CS6303 Large Language Models, CS331 Artificial Intelligence, CS487 Cloud Development, CS437 Deep Learning

## TECHNICAL SKILLS & INTERESTS

**Languages:** English, Urdu

**Technical Skills:** Python, MERN, Java, C++/C, MATLAB, Haskell, React, LaTeX, PyTorch, AWS

**Interests:** Formula 1, Golf, Horse Riding, Applied Machine Learning, Data Science

## RESEARCH EXPERIENCE

**Max Planck Institute SWS**

April 2026 - Present

Research Intern | Supervised by Dr. Abhilasha Ravichander

- Designing an LLM driven agentic system for inverse molecular design, combining generative models with structured tool-calling (RDKit, docking, QSAR) for multi-objective molecular optimization
- Building closed loop generation pipelines with iterative refinement and feedback driven reasoning, enabling improved validity, diversity, and binding affinity across candidate molecules

**Center for Speech and Language Technologies (CSaLT) – LUMS**

Sept 2024 – Present

Researcher | Supervised by Dr. Agha Ali Raza (LUMS) & Dr. Usman Naseem

- Developing self-evolving agentic frameworks with built-in cultural awareness, enhancing reasoning and tool-calling performance while improving robustness to Western-centric alignment biases
- Published Pakistan's first locally contextualized QA bias benchmark (PakBBQ) (17,180 QA pairs; 8 socio-cultural bias categories), benchmarking GPT-4.1, Gemini 2.0, and DeepSeek-V3. Demonstrated cross-linguistic bias disparities and up to ~12-point performance shifts.
- Led a team and developed Pak3H, a human validated Urdu benchmark suite for evaluating LLM alignment (helpfulness, harmlessness, honesty), exposing significant cross-lingual degradation in safety and truthfulness
- Researched 10+ Large Language Models (LLMs) for low-resource settings, analyzing bias vulnerabilities and jailbreak robustness, while optimizing speech/NLP models for Urdu language tasks.

**German Research Center for AI & RPTU**

July 2025 – March 2026

Lead Researcher | Supervised by Dr. Nabeel Asim (RPTU, DFKI)

- Developed a multi representation SMILES based Molecular Property Prediction Model using a multi-granular transformer framework leveraging atom-, functional-group-, and branch-level SMILES representations for robust molecular property prediction.
- Pretrained specialized **BERT**, **RoBERTa**, **DeBERTa** and **statespace** models on large scale SMILES corpora and distilled their knowledge into a unified student model using MLM + FitNet distillation with dynamic gating.
- Achieved **state-of-the-art** and consistent performance across **all 22 ADMET tasks**, outperforming models that excel only on task subsets.
- Built and processed a **500M+** compound dataset across **150+** molecular properties, leveraging large scale data aggregation, web scraping, and distributed GPU clusters for high throughput processing

**Confidence Aware Multi-Teacher Distillation for OOD Generalization (CITY, LUMS)** [Link](#)

Sept 2024 – Mar 2025

Primary Researcher | Supervised by Dr. Muhammad Tahir (LUMS)

- Proposed a confidence-aware multi-teacher knowledge distillation framework to mitigate texture/shape inductive biases and improve out-of-distribution (OOD) generalization in image classification useful in medical imaging
- Ensembled shape- and texture-biased ViT and VGG16 teachers using entropy-based logit weighting; trained a ViT-Small student via KL divergence with dynamic bias propagation.
- Achieved balanced cue reliance (Shape 0.49% / Texture 0.51%) and outperformed single-teacher and naïve KD baselines across ImageNet-1k, Stylized-ImageNet, and related OOD benchmarks, demonstrating improved robustness.

## RESEARCH PAPERS & PUBLICATIONS

**Pak3H- Evaluating the Cost of Cultural Mismatch in LLM Alignment (Under Review)** [Link](#)

Jan 2026 – March 2026

- Developed Pak3H, a human-validated Urdu benchmark suite of total 28000 samples for evaluating LLM alignment (helpfulness, harmlessness, honesty) in culturally grounded settings
- Designed a multi-stage annotation and localization pipeline ensuring semantic and cultural fidelity beyond translation based approaches
- Revealed systematic cross lingual alignment failures in state of the art LLMs, with consistent degradation across all 3H dimensions

**PakBBQ-A Culturally Adapted Bias Benchmark for QA (EMNLP 2025 Main Conference)** [Paper Link](#) Sept 2024 – May 2025

- Developed PakBBQ, the first culturally adapted QA benchmark (17,180 examples in Urdu and English across 8 sociocultural bias categories) to evaluate social bias in multilingual LLMs within the Pakistani context
- Evaluated 6 multilingual LLMs under zero shot conditions, revealing cross linguistic bias gaps and demonstrating that disambiguation and negative framing significantly improved bias mitigation and accuracy

TEACHING ASSISTANTSHIPS

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|-----------------------------------------------------------------------------|-----------------------------|
| • <b>Teaching Assistant</b> – <i>AI600 Machine Learning (Masters in AI)</i> | <i>Jan 2025 – May 2025</i>  |
| • <b>Teaching Assistant</b> – <i>CS5302 Foundations of Generative AI</i>    | <i>Jan 2025 – May 2025</i>  |
| • <b>Teaching Assistant</b> – <i>CS535 Machine Learning</i>                 | <i>Sept 2024 – Dec 2024</i> |
| • <b>Teaching Assistant</b> – <i>CS100 Computational Problem Solving</i>    | <i>Sept 2023 – Dec 2023</i> |

GITHUB PROJECTS

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| <b>Bias Mitigation in Deep Learning Vision Models</b> <a href="#">Link</a>                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Sept 2024 – Dec 2024</i> |
| <ul style="list-style-type: none"><li>Implemented a multi-teacher single-student knowledge distillation framework to address inductive biases in deep learning, leveraging CNNs for texture and Vision Transformers for shape representation</li><li>Designed space adapters for heterogeneous architectures, achieving a compact student model with superior generalisation on cue-conflict datasets under distributional shifts, resulting in <b>12%</b> less bias.</li></ul> |                             |
| <b>AI for All – Evaluating Transformer and LLM Models in Low Resourced Settings</b> <a href="#">Link</a>                                                                                                                                                                                                                                                                                                                                                                        | <i>Sept 2024 – Dec 2024</i> |
| <ul style="list-style-type: none"><li>Evaluated open source LLM and transformer model’s performance on low resource languages such as Urdu, Sindhi and Pashto. Tested jail breaking adversarial prompts with a success rate of up to <b>40%</b> in Llama-3B.</li></ul>                                                                                                                                                                                                          |                             |
| <b>AI-Driven Stock Insights Platform with AWS &amp; Claude Chatbot</b> <a href="#">Link</a>                                                                                                                                                                                                                                                                                                                                                                                     | <i>Sept 2024 – Dec 2024</i> |
| <ul style="list-style-type: none"><li>Built and deployed full-stack real-time stock analysis platform using AWS Lambda, S3, ECS, and Neon Postgres</li><li>Integrated Claude Haiku based chatbot with custom prompt chains for natural language queries and investment insights</li><li>Achieved <b>&lt;200ms</b> latency and auto-scaling to millions of users via serverless architecture and ECS Fargate</li></ul>                                                           |                             |
| <b>Agentic Data Science Chatbot</b> <a href="#">Link</a>                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Jan 2024 – May 2024</i>  |
| <ul style="list-style-type: none"><li>Built an agentic AI Data Science chatbot, aimed to perform intuitive statistical analysis, EDA and data cleaning, enhancing accessibility for users with limited Data Science knowledge</li><li>Fine tuned Llama 3 to analyse data in <b>10+</b> supported formats, deployed on Hugging Face using Gradio</li></ul>                                                                                                                       |                             |

INDUSTRY EXPERIENCE

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| <b>Darcheeni (AI in Healthcare Startup) – LUMS – MLOps Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>May 2024 – May 2025</i> |
| <ul style="list-style-type: none"><li>Led backend development for an AI-driven clinical assistant, integrating ASR, RL-based decision models, and LLM-driven response generation for real-time triage</li><li>Optimized speech recognition pipelines through data processing, model finetuning, and LLM-based post-processing for improved transcription quality in diverse settings</li><li>Designed and deployed serverless infrastructure on AWS (Lambda, DynamoDB, Auth0) with autoscaling, enabling scalable, low-latency model inference and system reliability</li></ul> |                            |
| <b>Delivery Hero (FoodPanda) – Sales Performance Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Jun 2024 – Aug 2024</i> |
| <ul style="list-style-type: none"><li>Conducted large-scale EDA on vendor onboarding and QC data to identify trends and bottlenecks, streamlining the self-signup process and reducing QC rejection rate by <b>15%</b>.</li><li>Developed an agentic workflow using Python and Google APIs to automatically pull daily data, generate invoices, and distribute them to vendors via WhatsApp, reducing manual effort and process time by <b>35%</b>.</li></ul>                                                                                                                   |                            |
| <b>Pakistan Television Network – Data Science Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Jun 2023 – Jul 2023</i> |
| <ul style="list-style-type: none"><li>Revamped and enhanced PTV World’s ticker headline system using machine learning models for text similarity, resulting in <b>20%</b> reduction in headline overheads</li><li>Built a predictive analysis model, enhancing viewer engagement metrics through data-driven content strategies by <b>10%</b></li></ul>                                                                                                                                                                                                                         |                            |

LEADERSHIP EXPERIENCE AND CO-CURRICULAR ACTIVITIES

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| <b>TechStars Startup Weekend, Lahore - Convenor</b>                                                                                                                                                                                                                                                                                                                                                                                        | <i>Jul 2022 – Mar 2023</i>  |
| <ul style="list-style-type: none"><li>Executed Startup Weekend Lahore 2023 in collaboration with National Incubation Center Lahore and Google for Startups, led workshops and partnered with top industry leaders for startup guidance and judging</li><li>Managed a 54-hour event for over <b>40</b> startups and <b>150+</b> attendees across Pakistan, overlooking a team of 60 and securing <b>800k+</b> PKR in sponsorships</li></ul> |                             |
| <b>SPADES - Assistant Director (Socials and Operations)</b>                                                                                                                                                                                                                                                                                                                                                                                | <i>Sept 2022 – May 2023</i> |
| <ul style="list-style-type: none"><li>Led the organisation of PSIFT's flagship event, coordinating social function and concerts for over <b>1,400</b> attendees</li></ul>                                                                                                                                                                                                                                                                  |                             |